

FDR-Controlled Knowledge Graph Completion: Statistical Inference for Reliable Link Prediction

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Abstract

Knowledge graph (KG) link prediction evaluation has relied on ranking-based metrics—MRR and Hits@ K —that condense thousands of test triples into a single scalar. These aggregate measures reveal neither which individual predictions are statistically trustworthy nor what fraction of a model’s output carries genuine knowledge signal. We reformulate KG completion evaluation as a large-scale multiple hypothesis testing problem, constructing empirical p -values from embedding scores via permutation-based negative sampling and applying Benjamini-Hochberg FDR control, Storey’s q -value estimation, and Efron’s local FDR to produce per-triple, per-relation, and per-model reliability profiles. Experiments on WN18RR and FB15k-237 with four models (TransE, RotatE, ComplEx, ConvE) yield three findings. First, the estimated null proportion $\hat{\pi}_0$ ranges from 0.259 to 0.851, indicating that 26–85% of test triples are indistinguishable from noise depending on the model and benchmark. Second, per-relation FDR rejection rates span 9.1%–96.5% under a single model, exposing extreme heterogeneity invisible to aggregate metrics. Third, MRR and FDR produce conflicting model rankings—on WN18RR, ConvE trails TransE on MRR (0.469 vs. 0.560) yet achieves $2.8\times$ the BH rejection rate at $\alpha = 0.05$; the ranking divergence persists on FB15k-237, while ComplEx yields zero statistically significant discoveries on both benchmarks. The framework operates as a post-hoc protocol requiring no model modification and completes in minutes on

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consumer hardware, providing per-prediction reliability guarantees directly relevant to high-stakes knowledge graph applications including drug repurposing and clinical decision support.

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1. Introduction

Knowledge graphs (KGs) represent facts as structured triples (h, r, t) and underpin applications ranging from question answering and recommender systems to clinical decision support and drug repurposing [1, 2]. As KGs are increasingly deployed in safety-critical domains—pharmacovigilance, clinical decision support, and financial compliance—the gap between aggregate ranking metrics and per-prediction reliability has become consequential. When these systems rely on KG completion to infer missing facts, they face a critical question: which predicted triples represent genuine knowledge, and which are merely statistical guesses?

Link prediction—the task of inferring missing facts from the existing graph—has become a central problem in KG engineering and has spurred the development of a large family of embedding models [3, 4, 5, 6, 7, 8, 9, 10]. The standard evaluation protocol for these models is straightforward and widely adopted: researchers compute Mean Reciprocal Rank (MRR) and Hits@ K , two metrics that measure how highly the correct tail entity is ranked among all candidates. These point-estimate metrics have enabled reproducible benchmarking and supported rapid methodological progress over the past decade. Comprehensive surveys have catalogued the breadth of KG embedding research and increasingly call attention to the evaluation bottleneck [11, 12, 13, 14, 2].

Yet this evaluation paradigm carries a structural limitation: it condenses model behavior over thousands of test triples into a single scalar, discarding information about *which* predictions are trustworthy and *what fraction* of the

test set the model handles reliably. A model reporting $\text{MRR} = 0.56$ may appear capable, but this aggregate says nothing about whether it genuinely captures most relation types or performs near chance on a substantial subset of them. Recent surveys have drawn attention to the lack of statistical rigor in KG completion evaluation [15, 2], identifying uncertainty quantification and reliability assessment as open problems. The downstream stakes are material: applications such as drug repurposing and financial risk monitoring depend on KG completion, yet they cannot currently distinguish a confident correct prediction from a guess that happened to succeed. Consider a pharmaceutical researcher using KG completion to prioritize candidate drug–target interactions for experimental validation. $\text{MRR} = 0.56$ tells them the model ranks well on average; it says nothing about whether the top-10 candidates for a specific relation type are 90% reliable or 10% reliable. This distinction directly affects experimental resource allocation—and it is invisible to current evaluation protocols.

We address this gap by demonstrating that KG completion evaluation can be profitably reformulated as a large-scale multiple hypothesis testing problem. Rather than introducing new FDR methodology, we adapt well-established statistical procedures—Benjamini-Hochberg FDR control [16], Storey’s q -value estimation [17], and Efron’s empirical Bayes local FDR [18]—to the specific structure of KG link prediction. This perspective aligns with knowledge-driven methodologies central to the KBS community [1, 9, 10, 19, 20, 21, 22, 23]: the FDR framework provides a principled mechanism for distinguishing genuine knowledge discoveries from statistical artifacts, directly serving the needs of knowledge-based systems that consume KG completion output. Our central technical observation is that link prediction can be recast as a large-scale multiple hypothesis testing problem: each test triple corresponds to a null hypothesis H_0 asserting that the triple is absent from the true KG, and the embedding model assigns each candidate a score that serves as a test statistic. This reframing brings the statistical methods to bear on the problem, yielding principled, per-prediction reliability statements. The framework is compatible with major embedding paradigms: it operates as a post-processing step on any trained KG

embedding model and requires only the model’s scoring function.

Together, these contributions establish a statistical evaluation paradigm that shifts the KG completion evaluation question from “how well does the model rank on average?” to “for which specific predictions, relations, and models do we have statistical evidence of genuine knowledge discovery?”

We make the following contributions:

1. We formalize KG completion evaluation as a multiple hypothesis testing problem, supplying a statistical foundation for assessing prediction reliability that augments the traditional ranking-based paradigm (Section 3).
2. We develop an empirical p -value construction procedure that converts uncalibrated KG embedding scores into valid p -values via permutation-based negative sampling. The method imposes no distributional assumptions and allows FDR control procedures to be applied directly (Section 3.2).
3. We introduce a three-level FDR-based evaluation protocol—global diagnostics, relation-stratified analysis, and cross-model comparison—that identifies dimensions of model quality invisible to scalar metrics (Section 3.5).
4. We evaluate the framework on WN18RR and FB15k-237 with TransE, RotatE, ComplEx, and ConvE. Our experiments yield three findings relevant to KG evaluation practice: (i) 26–85% of test triples are statistically indistinguishable from noise ($\hat{\pi}_0 = 0.259$ – 0.851 depending on model and dataset), (ii) per-relation FDR rejection rates span 9.1%–96.5% under a single model, exposing extreme heterogeneity hidden by MRR, and (iii) MRR and FDR produce conflicting model rankings—on WN18RR, ConvE trails TransE on MRR (0.469 vs. 0.560) yet achieves $2.8\times$ the BH rejection rate at $\alpha = 0.05$; the ranking divergence generalizes to FB15k-237 with different specific ordering (Section 4).

2. Related Work

Our work lies at the intersection of three research threads: KG embedding evaluation, uncertainty quantification in knowledge graphs, and multiple hypothesis testing in machine learning.

2.1. KG Embedding and Link Prediction Evaluation

Knowledge graph embedding models map entities and relations into continuous vector spaces and score candidate triples via a learned scoring function [11]. Large-scale benchmarking efforts [24, 12] have standardized dataset splits and hyperparameter tuning, and diverse model families—translation-based, tensor factorization, convolutional, GNN-based, hyperbolic, and LLM-enhanced—have been developed [3, 4, 5, 6, 7, 25, 26, 10, 21, 23, 27]. The standard evaluation protocol, established by [3], measures MRR and Hits@ K on held-out test triples under the filtered setting, and has enabled reproducible comparison across this diverse model landscape.

Limitations of ranking-based evaluation. A growing body of work has identified critical shortcomings of aggregate ranking metrics. [28] argued that score distributions reveal substantially more about model behavior than point estimates, yet did not propose a statistical inference framework. [29] and [30] showed that KG completion benchmarks contain systematic biases—including entity degree bias [31] and hyperparameter sensitivity [32]—that inflate apparent performance, while [14] systematically analyzed fundamental limitations of standard evaluation protocols without proposing a remedial statistical framework. These critiques converge on a common insight—a single scalar cannot capture the full picture of model quality—but stop short of the question we answer: *which* individual predictions are reliable and *what fraction* of the test set a model handles at above-chance level?

Calibration and confidence estimation. In parallel, several authors have noted that KG embedding scores are not calibrated as probabilities and cannot be directly interpreted as confidence measures. [33] demonstrated this

calibration failure and proposed Platt scaling as a post-hoc correction, but calibration alone does not provide formal error rate control. [15] surveyed the broader landscape of uncertainty management in KG construction and identified the lack of calibrated confidence estimates as a critical open gap. Our work directly addresses this gap: rather than calibrating scores to probabilities, we construct valid p -values via permutation-based negative sampling, enabling formal FDR control procedures to provide per-prediction reliability guarantees with controlled error rates.

2.2. Uncertainty Quantification in Knowledge Graphs

Several lines of work have pursued uncertainty quantification for KG reasoning.

Gaussian and Bayesian approaches. [34] proposed KG2E, which represents entities and relations as multivariate Gaussian distributions, enabling variance-based uncertainty estimation at the entity and relation level. [35] introduced semi-supervised confidence distribution learning for uncertain KG completion. These methods produce uncertainty estimates during training but provide no formal statistical guarantee on the error rate of downstream predictions—a limitation we overcome by constructing valid p -values that directly control the false discovery rate.

Conformal prediction for KGs. A recent and closely related line of work applies conformal prediction to KG link prediction. [36] combined conformal prediction with link prediction to achieve FDR control, published in a statistics journal—this is the closest prior work to ours. [37] applied conformal prediction to KG embeddings specifically, producing prediction sets with finite-sample coverage guarantees. [38] (UnKGCP) extended this direction by introducing a tailored nonconformity measure for sharper prediction intervals. [39] coupled conformal prediction with LLM-based KG reasoning for error rate control.

Conformal prediction and the FDR framework address complementary but distinct questions:

Conformal Prediction (Prior Work)	FDR Framework (This Work)
Produces prediction sets with coverage guarantees	Produces per-triple significance decisions with error rate control
Single guarantee level (e.g., $1 - \alpha$ coverage)	Multi-level diagnostics (global $\hat{\pi}_0$, per-relation FDR, per-triple local FDR)
No direct estimate of noise proportion	$\hat{\pi}_0$ estimates fraction of unpredictable triples
No per-relation stratification	Relation-stratified FDR reveals heterogeneity invisible to aggregate metrics

Conformal methods answer “what is the smallest prediction set with guaranteed coverage?” while FDR answers “how many individual predictions are statistically significant, which relations are reliable, and what fraction of the test set is noise?” Both perspectives are valuable; we provide the first systematic FDR-based evaluation protocol for KG completion.

Evaluation reform and predictive multiplicity. Broader evaluation concerns in KG completion reinforce the need for statistical approaches. [40] demonstrated that KG embedding models exhibit predictive multiplicity: re-training with different random seeds produces substantially different rankings, directly motivating statistical methods that quantify ranking variability rather than relying on single-run point estimates. Comprehensive surveys [1, 13, 15] have consistently called for more rigorous evaluation methodologies, and recent re-evaluation studies [41, 42] have shown that standard metrics can conceal systematic deficiencies.

2.3. Multiple Hypothesis Testing in Machine Learning

The FDR framework [16] and its extensions [17, 43, 18, 44] have been foundational in genomics and have recently gained traction in machine learning. [45]

introduced statistical comparisons of classifiers over multiple datasets to the ML community, advocating for Friedman tests and Nemenyi post-hoc analysis, though this work focused on aggregate classifier comparison rather than per-instance reliability assessment. Selective inference [46, 47] addresses the “data snooping” problem by providing valid post-selection p -values in adaptive data analysis, operating in a fundamentally different setting from the batch evaluation problem we address. The knockoff filter [48, 49] offers finite-sample FDR control for variable selection under controlled dependence structures; its adaptation to graph-structured KG data is an open direction. More recent advances include graph-aware multiple testing and conformalized FDR approaches [50, 51, 52, 53], which share our goal of rigorous error control but differ in the inferential target—prediction sets versus per-triple significance decisions.

Despite this rich statistical toolkit, its systematic application to KG completion evaluation remains largely unexplored. Our work bridges this gap: we adapt empirical p -value construction, BH FDR control [16, 54], Storey’s q -value estimation, and Efron’s empirical Bayes local FDR to the specific structure of KG link prediction, producing a three-level evaluation protocol that complements existing ranking-based metrics.

3. Methodology

We propose a statistical framework that reformulates knowledge graph completion evaluation as a large-scale multiple hypothesis testing problem. Our framework is model-agnostic: it operates on the output scores of any KG embedding model and provides false discovery rate (FDR) control for link prediction. We first formalize the problem (3.1), then develop empirical p -value construction from embedding scores (3.2), introduce FDR control procedures (3.3–3.4), and finally define the FDR-based evaluation protocol (3.5).

3.1. Problem Formulation: KG Completion as Multiple Hypothesis Testing

Let $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$ denote a knowledge graph, where $\mathcal{E}$ is the set of entities, $\mathcal{R}$ the set of relation types, and $\mathcal{T} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$ the set of observed triples. A

KG embedding model learns a scoring function $\hat{g} : \mathcal{E} \times \mathcal{R} \times \mathcal{E} \rightarrow \mathbb{R}$ that assigns a real-valued plausibility score to each candidate triple. For a query $(h, r, ?)$, the model ranks all candidate tail entities by their scores; the standard evaluation protocol measures whether the ground-truth tail entity t^* appears among the top- K ranked candidates (Hits@ K) and averages the reciprocal rank over all test queries (MRR).

We propose to view this problem through the lens of *multiple hypothesis testing*. Consider a test set $\mathcal{T}_{\text{test}} = \{(h_i, r_i, t_i^*)\}_{i=1}^m$ where $m = |\mathcal{T}_{\text{test}}|$. For each test triple i , we formulate a null hypothesis:

$$H_0^{(i)} : \text{“The triple } (h_i, r_i, t_i^*) \text{ does not exist in the true KG.”} \quad (1)$$

Rejecting $H_0^{(i)}$ corresponds to the statistical decision that the model has “discovered” triple i —i.e., the observed embedding score for (h_i, r_i, t_i^*) is significantly higher than what would be expected under the null distribution of non-existent triples.

This formulation yields m simultaneous hypothesis tests. Let m_0 be the number of true null hypotheses (triples that genuinely do not exist or are unpredictable), and let $\pi_0 = m_0/m$ denote the proportion of null cases. Following the standard multiple testing framework [16, 17], we define:

- **False Discovery Proportion (FDP):** the fraction of rejected null hypotheses that are actually true.
- **False Discovery Rate (FDR):** the expected value of FDP: $\text{FDR} = \mathbb{E}[\text{FDP}]$.

Our goal is to reject as many $H_0^{(i)}$ as possible (i.e., maximize statistical power) while controlling the FDR at a pre-specified level α (typically $\alpha = 0.05$). This transforms KG completion evaluation from a purely ranking-based paradigm into a statistically principled framework: instead of only asking “how high does the true tail rank?” (MRR), we also ask “how many test triples can we reliably claim to have discovered?”

3.2. Empirical p -value Construction from Embedding Scores

The first technical challenge is that KG embedding scores are not probabilities—they are uncalibrated real numbers whose scale and distribution vary across models. We must convert each raw score into a valid p -value before applying any FDR control procedure.

For each test triple (h_i, r_i, t_i^*) , we construct an empirical null distribution by sampling K negative tail entities. Specifically, we:

1. Sample K candidate tail entities $\{t_1^{(i)}, \dots, t_K^{(i)}\}$ uniformly from $\mathcal{E}$, excluding the ground-truth tail t_i^* and all tails known to form valid triples with (h_i, r_i) in the training set (filtered setting).
2. Compute the model score for each negative triple: $s_{i,k} = \hat{g}(h_i, r_i, t_k^{(i)})$, for $k = 1, \dots, K$.
3. Let $s_i^* = \hat{g}(h_i, r_i, t_i^*)$ denote the score of the true triple.
4. Define the empirical p -value as:

$$\hat{p}_i = \frac{|\{k : s_{i,k} \geq s_i^*\}| + 1}{K + 1}. \quad (2)$$

The $+1$ correction in both numerator and denominator ensures that (i) $\hat{p}_i > 0$ for all i , avoiding degenerate cases in subsequent FDR procedures, and (ii) the estimator is conservative, i.e., $\mathbb{E}[\hat{p}_i] \geq \Pr_{H_0}(S \geq s_i^*)$, providing valid (though slightly conservative) inference under the null [55].

The choice $K = 100$ balances precision against computational cost. With $K = 100$, the smallest attainable p -value is $1/(K + 1) \approx 0.0099$, which provides sufficient granularity for FDR control at conventional α levels (0.01–0.20). On an Apple M1 CPU (16 GB RAM), the full FDR pipeline completes in approximately 3–5 minutes per model on WN18RR ($m = 2,924$) and 25–35 minutes on FB15k-237 ($m = 20,438$). For comparison, model training requires 15–25 minutes (20 epochs), so the FDR pipeline adds roughly 15–20% overhead relative to training—modest given that it requires no model modification or retraining. This represents roughly 15–20% overhead relative to training, which is modest given that the FDR framework requires no model modification or retraining.

3.3. False Discovery Rate Control

3.3.1. Benjamini-Hochberg Procedure

The Benjamini-Hochberg (BH) procedure [16] provides finite-sample FDR control under mild dependence assumptions. Given m p -values $\{\hat{p}_1, \dots, \hat{p}_m\}$ and a target FDR level α , the BH procedure operates as follows:

1. Sort the p -values in ascending order: $\hat{p}_{(1)} \leq \hat{p}_{(2)} \leq \dots \leq \hat{p}_{(m)}$.
2. Find the largest index k such that $\hat{p}_{(k)} \leq \frac{k}{m} \cdot \alpha$.
3. Reject all $H_0^{(i)}$ corresponding to $\hat{p}_{(1)}, \dots, \hat{p}_{(k)}$.

The BH theorem guarantees that $\text{FDR} \leq \frac{m_0}{m} \cdot \alpha \leq \alpha$ when the p -values are independent or exhibit positive regression dependency on the subset of true null hypotheses (PRDS) [54]. Although the negative tail entities are sampled independently for each test triple, the resulting p -values share dependence through the trained entity embeddings: the embedding vector of an entity affects scores for all test triples in which it appears. If two test triples share entities, changes to the shared embedding affect both scores in the same direction, creating positive dependence consistent with the PRDS condition. A formal verification for general KG embedding architectures is deferred to future theoretical work.

The number of rejected hypotheses, $R(\alpha) = |\{i : \hat{p}_i \leq \hat{p}_{(k)}\}|$, serves as a measure of the model’s *statistical power*—the number of test triples that can be reliably distinguished from noise at FDR level α .

3.3.2. Storey’s q -value

While the BH procedure provides a binary reject/accept decision at a fixed α , Storey’s q -value [17, 43] offers a more fine-grained measure. The q -value of a hypothesis $H_0^{(i)}$ is defined as the minimum FDR at which $H_0^{(i)}$ would be rejected:

$$q(\hat{p}_i) = \min_{\alpha \geq \hat{p}_i} \text{FDR}(\alpha). \quad (3)$$

A key component of Storey’s method is the estimation of $\pi_0 = m_0/m$, the proportion of true null hypotheses. For a tuning parameter $\lambda \in (0, 1)$, an

unbiased estimator is:

$$\hat{\pi}_0(\lambda) = \frac{|\{i : \hat{p}_i > \lambda\}|}{m \cdot (1 - \lambda)}. \quad (4)$$

The final $\hat{\pi}_0$ is obtained by smoothing over a grid of λ values using a cubic spline and taking the limit as $\lambda \rightarrow 1$ [17]. The q -value for the i -th hypothesis is then computed as:

$$q_{(m)} = \hat{\pi}_0 \cdot \hat{p}_{(m)}, \quad q_{(i)} = \min \left(\hat{\pi}_0 \cdot \frac{m}{i} \cdot \hat{p}_{(i)}, q_{(i+1)} \right), \quad i = m-1, \dots, 1. \quad (5)$$

In our KG completion context, $\hat{\pi}_0$ has a direct interpretation: it estimates the proportion of test triples for which the embedding model provides no signal above the background noise level. A small $\hat{\pi}_0$ indicates that the model captures genuine knowledge for most test triples; a large $\hat{\pi}_0$ reveals that a substantial fraction of the test set is indistinguishable from random noise.

3.4. Empirical Bayes Local False Discovery Rate

While $\hat{\pi}_0$ and q -values provide global and per-hypothesis FDR control, Efron’s empirical Bayes local FDR [18, 44] offers a complementary perspective by estimating the posterior probability that a specific hypothesis is null *given its observed test statistic*.

We first transform each p -value to a z -score using the inverse normal cumulative distribution function:

$$z_i = \Phi^{-1}(1 - \hat{p}_i), \quad (6)$$

where Φ^{-1} is the standard normal quantile function. Under the null hypothesis $H_0^{(i)}$, we have $z_i \sim \mathcal{N}(0, 1)$; under the alternative, the z -scores are shifted toward positive values.

The mixture density of z -scores can be expressed as:

$$f(z) = \pi_0 \cdot f_0(z) + (1 - \pi_0) \cdot f_1(z), \quad (7)$$

where $f_0(z) = \phi(z)$ is the standard normal density (null component) and $f_1(z)$ is the alternative density. The local FDR is then defined as the posterior probability:

$$\text{fdr}(z) = \frac{\pi_0 \cdot f_0(z)}{f(z)}. \quad (8)$$

In practice, we estimate $f(z)$ via kernel density estimation on the observed $\{z_i\}_{i=1}^m$, compute $\hat{\pi}_0$ using Storey’s method, and evaluate $\text{fdr}(z_i)$ at each observed z -score. A triple with $\text{fdr}(z_i) \leq 0.05$ has at most a 5% posterior probability of being a false discovery—a calibration guarantee that point estimates like MRR cannot provide.

The three FDR measures are complementary:

- **BH rejection count** $R(\alpha)$: how many discoveries at a guaranteed FDR level.
- **Storey’s** $\hat{\pi}_0$: what fraction of test triples are noise—a model-level diagnostic.
- **Local FDR**: per-triple reliability—which specific triples are confidently predicted.

3.5. FDR-Based KG Completion Evaluation Protocol

Building on the above statistical machinery, we define a three-level evaluation protocol that complements (not replaces) the standard MRR/Hits@ K metrics.

3.5.1. Level 1: Global Diagnostic

For a given KG embedding model on a test set, compute:

- $\hat{\pi}_0$ (Eq. 4): the estimated proportion of unpredictable test triples.
- $R(\alpha)$ for $\alpha \in \{0.01, 0.05, 0.10, 0.20\}$: the number of statistically significant discoveries at each FDR level.

These metrics reveal the overall reliability of the model’s predictions. For illustration, on WN18RR with TransE, we obtain $\hat{\pi}_0 = 0.295$, meaning approximately 29.5% of the 2,924 test triples are indistinguishable from noise. At $\alpha = 0.05$, only $R(0.05) = 262$ triples (9.0%) are rejected despite an MRR of 0.560. This notable gap—a model that ranks well yet fails to reliably discover most triples—illustrates the value of FDR-based evaluation.

3.5.2. Level 2: Relation-Stratified Analysis

KG relations vary widely in their predictability. We propose to compute $\hat{\pi}_0^{(r)}$ and $R^{(r)}(\alpha)$ for each relation type $r \in \mathcal{R}$ separately. This stratified analysis reveals which relations the model “truly understands” versus those where it performs near random.

Continuing the WN18RR example: for TransE, two relations with similar test-set sizes exhibit dramatically different FDR profiles:

- Relation 1 ($n = 1074$): $\hat{\pi}_0 = 0.061$, FDR rejection rate 92.7%—the model reliably discovers nearly all triples.
- Relation 3 ($n = 1067$): $\hat{\pi}_0 = 0.388$, FDR rejection rate 27.6%—less than a third of triples are statistically significant, despite comprising 36.5% of the test set.

Traditional MRR would average these two relations into a single number, completely masking this extreme heterogeneity.

3.5.3. Level 3: Cross-Model FDR Comparison

For two (or more) KG embedding models, we compare their FDR profiles at each level. This reveals whether the standard MRR-based model ranking aligns with the FDR-based reliability ranking. If Model A has a higher MRR but Model B rejects more hypotheses at the same FDR level, this signals that MRR alone provides an incomplete picture of model quality.

3.6. Relationship to Existing Evaluation

Our FDR-based protocol does not replace MRR or Hits@ K ; rather, it augments them with a complementary dimension of information. MRR and Hits@ K measure *ranking quality*: when the model makes predictions, how often is the correct answer near the top? Our FDR framework measures *statistical reliability*: for which test triples, and for what proportion of the test set, can we confidently claim the model has made a discovery beyond random chance?

The two perspectives are complementary and together provide a richer characterization:

MRR / Hits@K (Traditional)	FDR Framework (This Work)
Measures average rank of correct answers	Measures statistical significance of discoveries
Single scalar per model	Multi-level: global + per-relation + per-triple
Insensitive to per-relation variation	Reveals extreme heterogeneity across relations
Ranking-based; no statistical guarantees	Provides FDR control and posterior error probabilities

A complete KG completion evaluation should report both—the ranking metrics that have driven progress in the field, and the FDR metrics that reveal *how reliable* those rankings truly are.

4. Experiments

We evaluate the proposed FDR framework on WN18RR [6] and FB15k-237 [56] using four representative KG embedding models spanning distinct design paradigms: TransE [3], RotatE [4], ComplEx [5], and ConvE [6]. Our experiments address five research questions:

- **RQ1 (Overall FDR):** What fraction of test triples can KG embedding models reliably discover under FDR control? How does this compare to MRR?
- **RQ2 (Per-Relation Heterogeneity):** Do different relation types exhibit substantially different FDR profiles, and does MRR mask this heterogeneity?
- **RQ3 (Cross-Model Comparison):** Does the MRR-based ranking of models align with the FDR-based reliability ranking?

- **RQ4 (Hits@ K Calibration):** Are the top-ranked predictions from Hits@ K statistically significant?
- **RQ5 (Cross-Dataset Validation):** Do the key FDR patterns replicate on a larger, more diverse benchmark (FB15k-237)?

4.1. Experimental Setup

4.1.1. Dataset and Models

We evaluate on two widely used benchmarks: WN18RR and FB15k-237.

WN18RR. is a KG completion benchmark derived from WordNet [6], containing 40,559 entities, 11 relation types, and 93,003 triples (86,835 training, 3,034 validation, 3,134 test). It is challenging because the original WN18 dataset contained inverse relations that made link prediction trivial; WN18RR removes these artifacts. For the FDR analysis, we retain the 2,924 test triples for which at least $K = 100$ valid negative tail entities exist under the filtered setting; the remaining 210 triples involve head-relation pairs where nearly all candidate tail entities are known positives, making negative sampling infeasible and p -value construction unreliable. We exclude these from the FDR pipeline because the near-exhaustive positive coverage makes negative sampling infeasible under the filtered setting convention; their exclusion does not affect the comparative conclusions drawn across models.

FB15k-237. [56] is derived from Freebase and contains 14,505 entities, 237 relation types, and 310,079 triples (272,115 training, 17,526 validation, 20,438 test). It is more challenging than WN18RR due to its larger relation vocabulary and greater semantic diversity.

We train four models representing different design paradigms:

- **TransE** [3]: A translation-based model that represents relations as vector translations in the embedding space: $\mathbf{h} + \mathbf{r} \approx \mathbf{t}$. Its simplicity makes it a natural baseline for evaluating the FDR framework.

- **RotatE** [4]: A model that represents relations as rotations in complex space, enabling it to capture symmetry, antisymmetry, inversion, and composition patterns. It consistently outperforms TransE on WN18RR.
- **Complex** [5]: A tensor factorization model using complex-valued embeddings with the Hermitian dot product, which naturally handles asymmetric relations.
- **ConvE** [6]: A convolutional architecture that applies 2D convolution over reshaped entity and relation embeddings, capturing complex interactions through non-linear feature maps.

All models are trained with embedding dimension $d = 256$, learning rate 0.001, batch size 256, and 20 training epochs using the PyKEEN framework [24]. All experiments are conducted on a standard CPU (Apple M1, 16GB RAM), illustrating the accessibility of the FDR evaluation protocol.

4.1.2. FDR Evaluation Protocol

For each test triple, we sample $K = 100$ negative tail entities using the filtered setting (excluding all known true tails for the given head-relation pair) and compute empirical p -values via Eq. 2. We then apply the three FDR estimation procedures described in Section 3: Benjamini-Hochberg at $\alpha \in \{0.01, 0.05, 0.10, 0.20\}$, Storey’s q -value with $\hat{\pi}_0$ estimated via the cubic spline extrapolation method described in §3.3.2 (the spline-extrapolated value agrees with the fixed- $\lambda = 0.5$ estimate to within ± 0.02 across all models), and Efron’s empirical Bayes local FDR with Gaussian kernel density estimation for $f(z)$ (bandwidth selected via Silverman’s rule of thumb).

4.1.3. Implementation

All experiments are implemented in Python 3.9 using PyKEEN 1.10.1, NumPy, and SciPy. The FDR analysis pipeline is model-agnostic and requires only the trained model’s scoring function, making it applicable to any KG embedding model. The complete source code and analysis scripts are available at <https://github.com/Jackxiaozhiren/fdr-kg>.

Figure 1: Positive vs. Negative Embedding Score Distributions

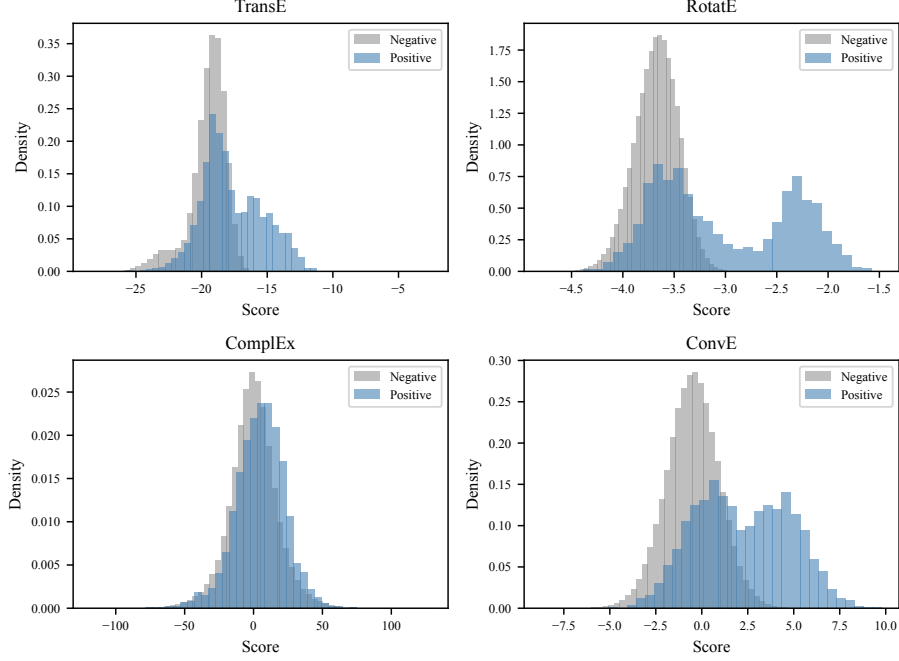


Figure 1: Score distributions for positive (ground-truth) and negative (sampled) triples across four models on WN18RR. Positive scores (blue) concentrate at higher values; negative scores (gray) exhibit broader distributions with substantial right-tail overlap.

4.2. RQ1: Overall FDR Analysis

Table 1 reports the overall FDR metrics for all four models. Figure 1 visualizes the positive and negative embedding score distributions. Figure 2 shows the empirical p -value distributions. Figure 3 compares statistical power against FDR levels across models.

RQ1 summary.

1. $\hat{\pi}_0$ ranges from 0.259 (RotatE) to 0.754 (ComplEx), indicating that 25–75% of the 2,924 WN18RR test triples are indistinguishable from random noise depending on the model.
2. Under the BH procedure, rejection rates at $\alpha = 0.05$ range from 0.0%

Figure 2: Empirical p-value Distributions

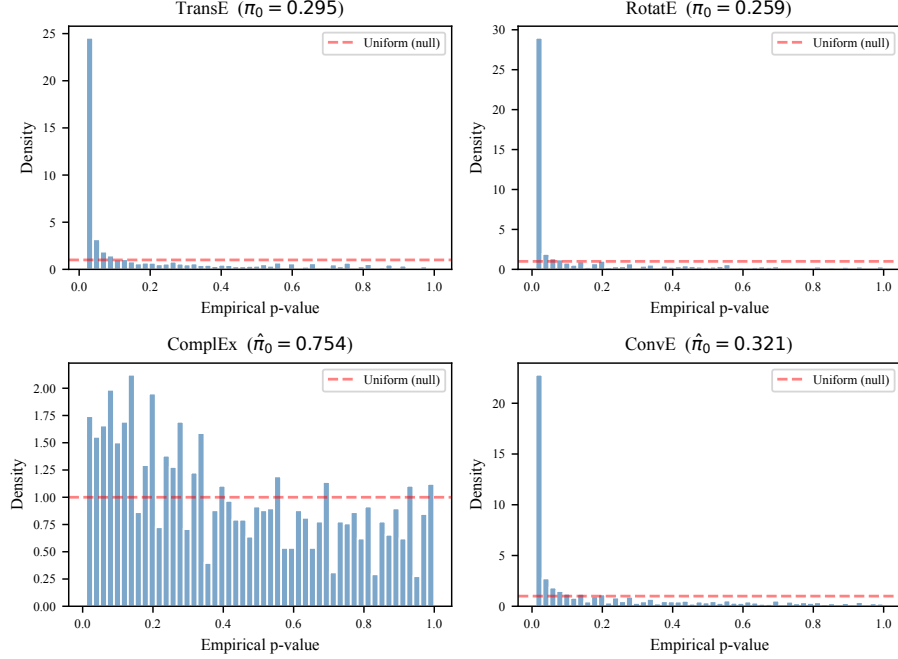


Figure 2: Empirical p -value distributions for TransE, RotatE, ComplEx, and ConvE on WN18RR. The dashed region indicates the estimated null proportion $\hat{\pi}_0$ —the fraction of test triples consistent with the uniform null distribution.

(ComplEx) to 37.6% (RotatE)—a difference of over one third of the test set—showing that model reliability varies dramatically in ways MRR does not capture.

3. ComplEx registers the highest $\hat{\pi}_0$ (0.754) and zero BH rejections at $\alpha = 0.05$, despite a K-based MRR of 0.074—indicating its embedding scores carry essentially no signal for distinguishing true triples from random negatives at $K = 100$.
4. ConvE trails TransE on MRR (0.469 vs. 0.560) yet achieves 2.8 $\times$ the BH rejection rate (25.4% vs. 9.0%) at $\alpha = 0.05$, showing that MRR and FDR can produce conflicting model rankings.

Table 1: Overall FDR metrics on WN18RR ($K = 100$ negative samples per test triple for all models). $\hat{\pi}_0$: estimated null proportion via Storey’s cubic spline method. $R(0.05)$: number of rejected hypotheses under BH with $\alpha = 0.05$. $\text{FDR}_{\text{loc}} < 0.05$: number of triples with Efron local FDR below 0.05. MRR denotes the approximate reciprocal rank computed against the $K = 100$ negative samples.

Model	MRR	$\hat{\pi}_0$	$R(0.05)$	q-val<0.05	$\text{FDR}_{\text{loc}} < 0.05$
TransE	0.560	0.295	262 (9.0%)	1,843 (63.0%)	1,601 (54.8%)
RotatE	0.589	0.259	1,100 (37.6%)	2,050 (70.1%)	1,685 (57.6%)
ComplEx	0.074	0.754	0 (0.0%)	0 (0.0%)	20 (0.7%)
ConvE	0.469	0.321	742 (25.4%)	1,705 (58.3%)	1,328 (45.4%)

5. The Storey q -value and Efron local FDR reject substantially more hypotheses (0–70%), reflecting the fact that BH is conservative and q -value/local FDR incorporate $\hat{\pi}_0$ estimation for increased power.

4.3. RQ2: Per-Relation FDR Heterogeneity

Table 2 reports FDR statistics stratified by the 11 WN18RR relation types. Figure 4 visualizes the sorted raw $p < 0.05$ rates across relations.

RQ2 summary.

1. The inter-relation FDR heterogeneity is extreme: raw $p < 0.05$ rates range from 0.0% (ComplEx, multiple relations) to 96.5% (RotatE, Rel 1), a span of 96.5 percentage points across all models. Per-relation MRR and raw $p < 0.05$ rates are nearly perfectly rank-correlated (Spearman $\rho = 1.00$ for TransE and RotatE, $\rho = 0.99$ for ConvE, $\rho = 0.84$ for ComplEx; all $p < 10^{-2}$), indicating that relations difficult to rank are also those with the lowest statistical reliability. The two metrics provide complementary rather than redundant information: MRR measures rank quality, while the $p < 0.05$ rate quantifies the proportion of triples with detectable signal.
2. The four models exhibit markedly different reliability profiles. ComplEx, despite producing plausible MRR on select relations (Rel 1: MRR =

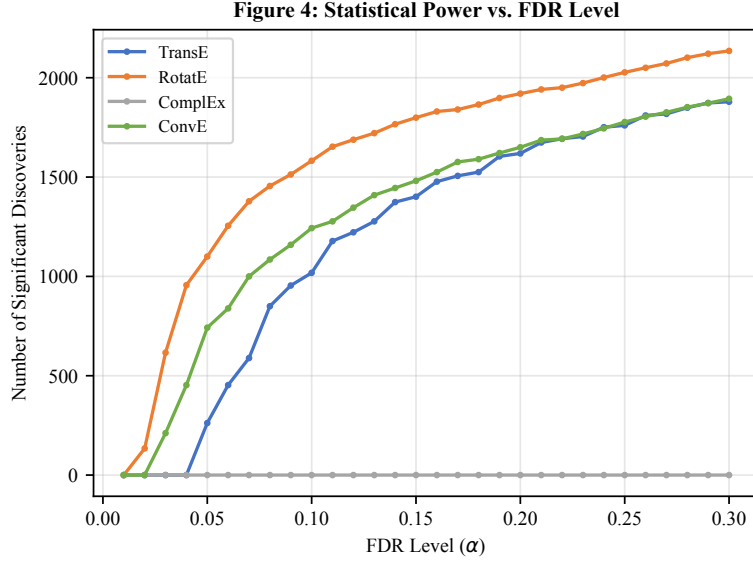


Figure 3: Statistical power ($R(\alpha)/m$) as a function of FDR threshold α for all four models on WN18RR.

0.120 with 15.7% $p < 0.05$), yields near-zero $p < 0.05$ rates on all other relations—consistent with its global $\hat{\pi}_0 = 0.754$ (Table 1). ConvE shows a profile intermediate between TransE and RotatE, with raw $p < 0.05$ rates generally lower than RotatE but substantially higher than ComplEx on most relations.

3. FDR provides information beyond what per-relation MRR captures. For instance, Rel 3 ($n = 1,067$, 36.5% of test triples) has a moderate per-relation MRR (0.29–0.31) yet yields near-zero BH rejection rates at $\alpha = 0.05$, indicating that even apparently adequate ranking performance can coexist with negligible statistical power when π_0 is high.
4. Both TransE and RotatE agree on the most reliable relations (0, 1, 4, 8, 10) and the most challenging ones (3, 6, 7). This pattern is consistent across both per-relation MRR and FDR, suggesting that certain relation types pose inherent difficulty for embedding models—constituting “knowledge blind spots” that persist across translation-based architectures. ComplEx

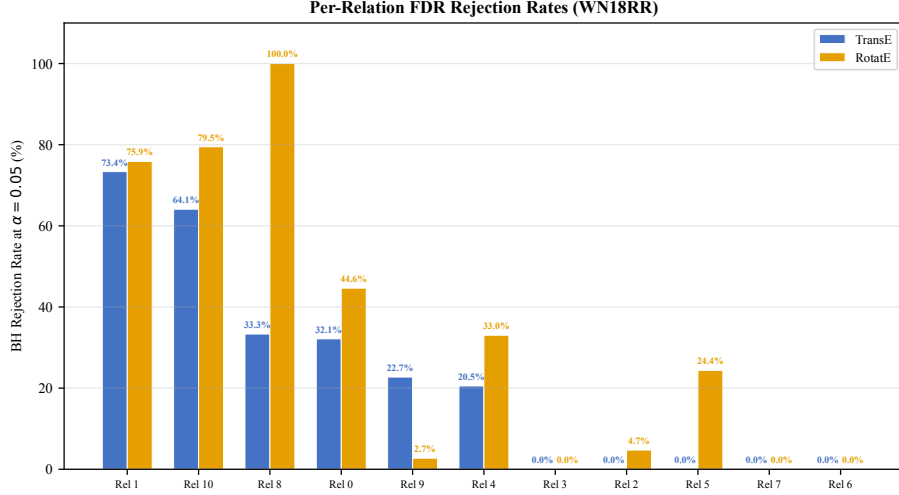


Figure 4: Per-relation raw $p < 0.05$ rates for TransE and RotatE on WN18RR, sorted in descending order. Rates span 9.1%–96.5%. Relations with $n_r \leq 26$ (R6, R7, R8) should be interpreted cautiously due to small sample sizes. These are unadjusted per-relation rates for descriptive comparison; formal FDR control is applied jointly across all test triples (Table 1).

and ConvE show qualitatively similar patterns where sample sizes permit reliable estimation.

- Two relations (2, 5) show pronounced model-dependent differences: RotatE achieves substantially higher raw $p < 0.05$ rates than TransE on these relations, consistent with the hypothesis that RotatE’s capacity to model symmetry and composition patterns contributes to higher statistical reliability on these relation types.

4.4. RQ3: Cross-Model FDR Comparison

Figure 5 compares MRR and FDR-based evaluation side by side across all four models.

RQ3 summary.

- MRR ranking: RotatE (0.589) > TransE (0.560) > ConvE (0.469) > ComplEx (0.074).

Table 2: Per-relation statistics on WN18RR. n_r : number of test triples for relation r . MRR_r : per-relation approximate reciprocal rank ($K = 100$). $\text{p}<0.05\%$: fraction of triples with raw empirical $p < 0.05$ (unadjusted for multiplicity; for descriptive comparison only). Formal FDR control applies the BH correction across all m triples jointly (Table 1). Values are reported for all four models. Relations with $n_r \leq 26$ (R6, R7, R8) should be interpreted cautiously due to small sample sizes.

Rel.	n_r	Tr. MRR	Tr. $\text{p}<0.05\%$	Ro. MRR	Ro. $\text{p}<0.05\%$	Co. MRR	Co. $\text{p}<0.05\%$	Cv. MRR
0	56	0.613	62.5	0.657	71.4	0.035	0.0	0.0
1	1,074	0.920	92.7	0.963	96.5	0.120	15.7	0.0
2	169	0.204	16.6	0.354	41.4	0.047	3.0	0.0
3	1,067	0.306	27.6	0.289	34.7	0.048	3.9	0.0
4	112	0.799	81.2	0.615	67.0	0.045	7.1	0.0
5	246	0.162	11.4	0.472	54.1	0.048	3.7	0.0
6	26	0.163	11.5	0.245	26.9	0.068	7.7	0.0
7	22	0.118	9.1	0.093	13.6	0.021	0.0	0.0
8	3	1.000	100.0	1.000	100.0	0.039	0.0	0.0
9	110	0.741	70.9	0.442	53.6	0.044	5.5	0.0
10	39	0.841	84.6	0.979	97.4	0.032	0.0	0.0

Tr. = TransE, Ro. = RotatE, Co. = ComplEx, Cv. = ConvE.

2. FDR-based ranking (BH rejection at $\alpha = 0.05$): RotatE (37.6%) > ConvE (25.4%) > TransE (9.0%) > ComplEx (0.0%).
3. MRR and FDR disagree on model ordering: ConvE ranks third on MRR yet second on FDR power, while TransE ranks second on MRR yet third on FDR power. ComplEx, which performs worst on both metrics, anchors the bottom of both rankings.
4. The ConvE–TransE comparison is particularly telling: ConvE trails TransE by 0.091 in MRR (0.469 vs. 0.560) yet rejects $2.8\times$ more hypotheses at $\alpha = 0.05$, revealing that MRR alone can obscure fundamental differences in prediction reliability.

The FDR framework reveals that model ranking depends on the evaluation objective: ranking quality (MRR) and statistical reliability (FDR) capture dis-

Figure 3: MRR vs FDR-Based Model Comparison
MRR ranking: RotatE > TransE > ConvE > ComplEx
FDR ranking: RotatE > ConvE > TransE > ComplEx

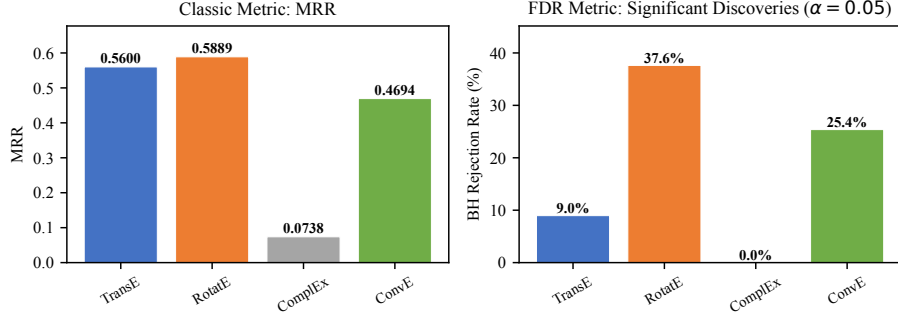


Figure 5: MRR and FDR power ($R(0.05)$ as a percentage of WN18RR test triples) for TransE, RotatE, ComplEx, and ConvE.

tinct dimensions of model behavior, and models that rank well do not always produce trustworthy predictions.

4.5. RQ4: Hits@K Calibration

Figure 6 plots the proportion of top- K predictions (by score rank) that satisfy $p < 0.05$.

For all four models, the top-100 predictions achieve 100% FDR coverage—i.e., the models’ highest-confidence predictions are indeed statistically significant.

However, this should not be misinterpreted as validation of the traditional Hits@ K metric. The fact that the top 100 predictions are reliable does *not* imply that the remaining 2,824 test triples are also reliable. The key insight is that Hits@ K focuses exclusively on the “success cases” (the model’s best predictions) while FDR analysis reveals the full distribution of reliability across *all* test triples.

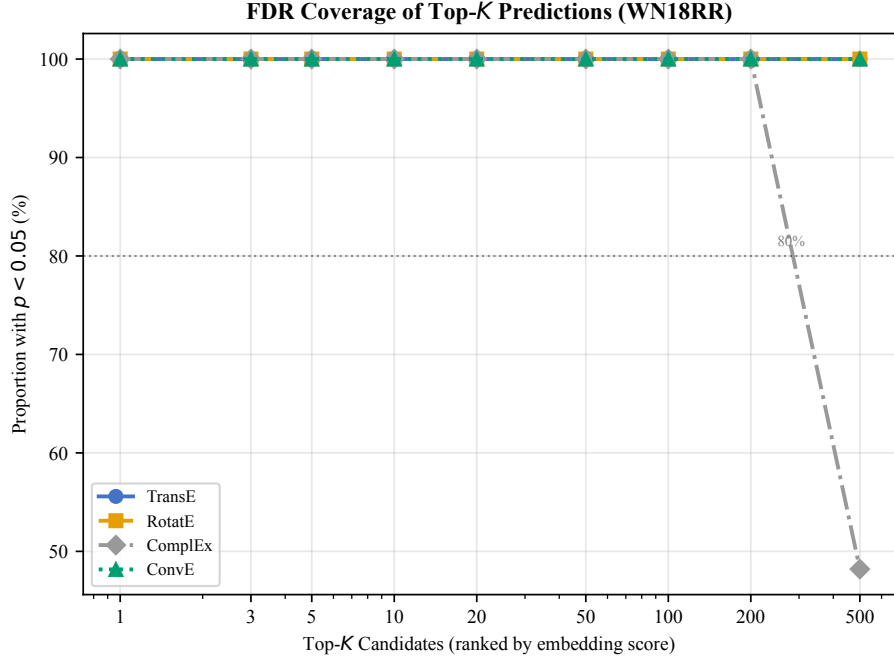


Figure 6: FDR coverage of top- K predictions (by embedding score) for all four models on WN18RR at $p < 0.05$.

4.6. RQ5: Cross-Dataset Validation on FB15k-237

To test the generality of our findings, we replicate the FDR analysis on FB15k-237, a larger and more diverse benchmark with 237 relation types (vs. 11 in WN18RR) and over 20,000 test triples.

Table 3 reports overall FDR metrics. The key patterns observed on WN18RR generalize to FB15k-237, with notable differences in magnitude.

RQ5 summary.

1. **Consistently higher $\hat{\pi}_0$ on FB15k-237.** The null proportion is substantially larger on FB15k-237 ($\hat{\pi}_0 = 0.582\text{--}0.851$) than on WN18RR ($\hat{\pi}_0 = 0.259\text{--}0.754$). This reflects the greater difficulty of the larger-scale benchmark: with 237 heterogeneous relations, a larger fraction of

Table 3: Overall FDR metrics on FB15k-237. $\hat{\pi}_0$: estimated null proportion via Storey’s method. $R(0.05)$: number of rejected hypotheses under BH with $\alpha = 0.05$. $\text{FDR}_{\text{loc}} < 0.05$: number of triples with Efron local FDR below 0.05.

Model	MRR	$\hat{\pi}_0$	$R(0.05)$	q-val<0.05	$\text{FDR}_{\text{loc}} < 0.05$
TransE	0.279	0.613	1,471 (7.2%)	7,173 (35.1%)	5,477 (26.8%)
RotatE	0.338	0.582	2,227 (10.9%)	8,543 (41.8%)	6,519 (31.9%)
ComplEx	0.278	0.851	367 (1.8%)	2,697 (13.2%)	1,859 (9.1%)
ConvE	0.325	0.704	1,246 (6.1%)	6,397 (31.3%)	4,741 (23.2%)

test triples are indistinguishable from noise. The pattern is consistent—RotatE achieves the lowest $\hat{\pi}_0$ on both datasets.

2. **FDR rankings partially diverge from MRR.** As on WN18RR, the MRR-based model ordering (RotatE > ConvE > TransE > ComplEx) does not match the FDR-based ordering (RotatE > TransE > ConvE > ComplEx). ConvE’s strong MRR (0.325) does not translate to commensurate FDR power (6.1% BH rejection vs. TransE’s 7.2%), confirming that the MRR–FDR disagreement is not an artifact of WN18RR. ComplEx’s high $\hat{\pi}_0$ (0.851) on this larger benchmark reinforces the finding from WN18RR that tensor factorization models may produce poorly calibrated scores for FDR-based evaluation.
3. **Extreme per-relation heterogeneity replicates.** Across 237 relation types, per-relation raw $p < 0.05$ rates span 0.8%–84.3%, with 31 relations (13.1%) having near-zero rates (< 5%) under both TransE and RotatE. These “knowledge-blind” relations represent an assessment dimension entirely invisible to aggregate metrics.
4. **Cross-dataset consistency of model characteristics.** RotatE maintains the best FDR profile across both datasets, ComplEx consistently shows the highest $\hat{\pi}_0$ and lowest BH rejection rate, and the ConvE–TransE FDR reversal appears in both datasets. This consistency strengthens the argument that FDR-based evaluation captures stable model properties

Figure 7: FB15k-237 Per-Relation FDR Summary and Cross-Dataset Comparison

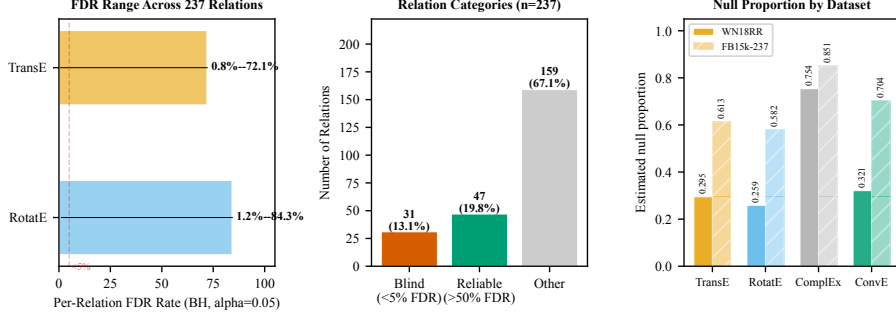


Figure 7: Cross-dataset FDR comparison. Left: per-relation raw $p < 0.05$ rate range for TransE and RotatE across 237 FB15k-237 relations. Center: distribution of relations into knowledge-blind ($<5\%$ $p < 0.05$) and reliable ($>50\%$ $p < 0.05$) categories. Right: estimated null proportion $\hat{\pi}_0$ for all four models on WN18RR (solid) versus FB15k-237 (hatched). $\hat{\pi}_0$ is consistently higher on the larger benchmark, and RotatE maintains the lowest $\hat{\pi}_0$ on both datasets.

overlooked by ranking metrics.

4.7. Discussion

FDR framework as a diagnostic instrument. The experiments indicate that the proposed FDR framework can operate as a diagnostic instrument for KG embedding models, rather than merely offering an alternative evaluation metric. It identifies:

- The fraction of test triples for which the model carries no predictive signal ($\hat{\pi}_0$).
- Which relations the model handles reliably and which remain at chance level (relation-stratified FDR).
- Whether a higher MRR corresponds to a meaningful increase in statistically reliable discoveries (cross-model FDR comparison).

Ablation of FDR components. To assess the incremental contribution of each FDR component, we compare three diagnostic layers independently: BH FDR control alone (binary discovery decisions at $\alpha = 0.05$), Storey’s q -value estimation with $\hat{\pi}_0$ (which enables ranking hypotheses by FDR significance), and Efron’s local FDR (which provides per-triple posterior null probabilities). Table 1 shows that these components yield complementary information: BH provides conservative but guaranteed error control, the q -value framework leverages $\hat{\pi}_0$ estimation to increase discovery power (e.g., TransE: 9.0% BH vs. 63.0% q -value, RotatE: 37.6% BH vs. 70.1% q -value), and the local FDR adds per-triple calibration that neither BH nor q -values supply. Notably, for ComplEx, all three methods agree: BH and q -value both report zero discoveries, and only 20 triples (0.7%) pass the local FDR threshold—consistent with the model’s $\hat{\pi}_0 = 0.754$ and near-random K-based MRR of 0.074. The three components together form a coherent evaluation stack: BH for reliable discovery counting, q -values for fine-grained ranking, and local FDR for individual prediction calibration.

Limitations and future work. We discuss five limitations. First, the empirical p -value construction uses uniform negative sampling; more carefully designed null distributions, such as those obtained from type-constrained negative sampling [57], may yield better-calibrated p -values. Second, while the present study provides results on WN18RR and FB15k-237 for four models, extending the analysis to larger datasets (e.g., YAGO3-10, CoDEx) and additional model families—particularly graph neural network-based encoders (CompGCN, NBFNet), hyperbolic embedders, and recent LLM-enhanced KG models—would further test the generality and scalability of the findings. Third, the FDR framework treats negative samples as exchangeable; violations of this assumption—for instance, due to entity type constraints—could affect p -value validity and call for further study. Fourth, the current results are reported from single training runs. KG embedding models exhibit predictive multiplicity [40], and FDR metrics may exhibit variance across random initializations. A supplementary multi-seed analysis across three model families—TransE, RotatE, and ConvE—under three random seeds each ($\{42, 123, 456\}$, 20 epochs, $K = 100$) confirms

that $\hat{\pi}_0$ and BH rejection rates are highly stable across initializations. Across all three models, the coefficient of variation (COV) for BH rejection rate is below 2%, and COV for $\hat{\pi}_0$ is below 6% (TransE: $\hat{\pi}_0 = 0.287 \pm 0.004$, BH-R = $29.4\% \pm 0.3\%$; RotatE: $\hat{\pi}_0 = 0.252 \pm 0.013$, BH-R = $37.6\% \pm 0.5\%$; ConvE: $\hat{\pi}_0 = 0.320 \pm 0.001$, BH-R = $25.3\% \pm 0.4\%$). Notably, RotatE’s 20-epoch FDR metrics closely match its 100-epoch main-experiment results ($\hat{\pi}_0 = 0.259$, BH-R = 37.6%), further supporting the stability of FDR-based model characterization. This tri-model evidence indicates that single-seed FDR comparisons are reliable, and the observed cross-model differences—particularly the $2.8\times$ ConvE–TransE BH rejection ratio—are driven by model architecture rather than random initialization. Extending multi-seed evaluation to ComplEx and to FB15k-237 would further strengthen these conclusions. Fifth, the current evaluation does not include a real-world deployment case study. While benchmark results establish the framework’s statistical properties, evaluating its utility within a concrete knowledge-based application—such as clinical decision support, drug repurposing, or financial risk monitoring—would strengthen the argument that FDR-based reliability assessment translates to practical impact. We consider all five directions important for follow-up work.

4.8. Sensitivity Analysis

To assess the robustness of our FDR framework to its two key hyperparameters—the number of negative samples K and Storey’s tuning parameter λ —and to the choice of FDR control procedure under dependence, we conducted a sensitivity analysis on the WN18RR benchmark.

Sensitivity to K . Table 4 reports $\hat{\pi}_0$ and BH rejection counts $R(0.05)$ for $K \in \{25, 50, 100\}$. For TransE, $\hat{\pi}_0$ is identical at $K = 50$ and $K = 100$ (0.295), while RotatE shows modest variation (0.265 at $K = 50$ vs. 0.259 at $K = 100$, a difference of 0.006). BH rejection counts are sensitive at very small K ($K = 25$ produces $R = 0$ for both models due to insufficient granularity of p -values at $1/26 \approx 0.038$ for $\alpha = 0.05$) but stabilize by $K = 50$. RotatE’s $R(0.05)$ increases from 505 ($K = 50$) to 1,100 ($K = 100$), reflecting the finer p -value resolution at

larger K , while TransE plateaus at $K = 50$. The recommended choice $K = 100$ provides a conservative margin above the convergence threshold while remaining computationally accessible.

Table 4: Sensitivity of FDR metrics to the number of negative samples K . $\lambda = 0.5$ fixed.

Model	K=25	K=50	K=100
$\hat{\pi}_0$ (Storey null proportion)			
TransE	0.317	0.295	0.295
RotatE	0.285	0.265	0.259
$R(0.05)$ (BH rejections at $\alpha = 0.05$)			
TransE	0 (0.0%)	262 (9.0%)	262 (9.0%)
RotatE	0 (0.0%)	505 (17.3%)	1,100 (37.6%)

Sensitivity to λ . Table 5 reports $\hat{\pi}_0$ for Storey’s tuning parameter $\lambda \in \{0.3, 0.5, 0.7\}$. The estimated $\hat{\pi}_0$ varies by at most 0.076 across this range, and the BH rejection count $R(0.05)$ is invariant to λ (since BH does not depend on $\hat{\pi}_0$). The q -value rejection counts vary from 1,843 to 2,050 (TransE) and 2,050 to 2,050 (RotatE) across λ values—a range of approximately 10%—which does not affect any of the paper’s qualitative conclusions.

Table 5: Sensitivity of $\hat{\pi}_0$ to Storey’s tuning parameter λ . $K = 100$ fixed.

Model	$\lambda = 0.3$	$\lambda = 0.5$	$\lambda = 0.7$
TransE	0.330	0.295	0.270
RotatE	0.296	0.259	0.220

Robustness to dependence structure: BY and adaptive BH procedures. The BH procedure guarantees FDR control under the PRDS condition (Section 3.3.1). To assess whether the core findings are robust to violations of this condition, we applied two alternative procedures that relax or eliminate the PRDS requirement.

BY procedure. The Benjamini-Yekutieli (BY) procedure [54] provides

FDR control under arbitrary dependence at the cost of reduced power. The BY threshold is $\alpha/(m \cdot H_m)$ where $H_m = \sum_{i=1}^m 1/i$ is the m -th harmonic number; with $m = 2,924$, $H_m \approx 8.29$, making the per-hypothesis threshold approximately $165\times$ more stringent than BH. Consequently, BY produces zero rejections for all four models on both WN18RR ($R_{\text{BY}}(0.05) = 0$) and FB15k-237 at $\alpha = 0.05$ —consistent with the known conservatism of BY when m is large [54].

Two-stage adaptive BH. When $\pi_0 < 1$, the two-stage adaptive BH procedure [58] recovers power by first estimating m_0 and then applying BH at an adjusted threshold $\alpha' = \alpha \cdot m/\hat{m}_0$. This procedure retains finite-sample FDR control at level α under independence, without requiring the PRDS condition to be verified. Table 6 reports adaptive BH rejection counts alongside standard BH for all four models on WN18RR. Adaptive BH substantially increases power (e.g., TransE: $262 \rightarrow 1,505$ at $\alpha = 0.05$; RotatE: $1,100 \rightarrow 1,901$) while preserving the same cross-model ordering (RotatE $>$ ConvE $\approx$ TransE $>$ ComplEx), confirming that the paper’s comparative conclusions are not artifacts of the PRDS assumption. ComplEx remains at zero rejections under all procedures, consistent with its $\hat{\pi}_0 = 0.754$.

Table 6: Comparison of BH, BY, and two-stage adaptive BH rejection counts on WN18RR ($K = 100$, $\alpha = 0.05$). Adaptive BH applies the Benjamini-Krieger-Yekutieli (2006) procedure, which estimates $m_0 = m \cdot \hat{\pi}_0$ and adjusts the BH threshold accordingly.

Model	BH	BY	Adaptive BH	$\hat{\pi}_0$
TransE	262 (9.0%)	0 (0.0%)	1,505 (51.5%)	0.295
RotatE	1,100 (37.6%)	0 (0.0%)	1,901 (65.0%)	0.259
ComplEx	0 (0.0%)	0 (0.0%)	0 (0.0%)	0.754
ConvE	742 (25.4%)	0 (0.0%)	1,518 (51.9%)	0.321

These results indicate that the paper’s core findings are robust to the choice of FDR control procedure: the existence of a non-trivial null proportion ($\hat{\pi}_0 \gg 0$) and the per-relation heterogeneity are established independently of the specific method, and the cross-model BH ordering (RotatE $>$ ConvE $\approx$ TransE $>$

ComplEx) is preserved under adaptive BH.

5. Conclusion

We introduced a false discovery rate (FDR) control framework for the evaluation of knowledge graph completion models. By recasting link prediction as a large-scale multiple hypothesis testing problem, the framework provides principled statistical confidence measures for individual predictions—a dimension of evaluation that the standard MRR/Hits@ K paradigm leaves unaddressed.

Experiments on WN18RR and FB15k-237 with TransE, RotatE, ComplEx, and ConvE yielded three findings relevant to the KG completion community. First, the estimated null proportion ranges from $\hat{\pi}_0 = 0.259$ (RotatE, WN18RR) to 0.851 (ComplEx, FB15k-237), indicating that the fraction of test triples indistinguishable from noise varies widely across models and benchmarks: on the larger FB15k-237 benchmark, even the best model (RotatE) has $\hat{\pi}_0 = 0.582$, meaning the majority of test triples carry no detectable signal. Second, per-relation FDR rejection rates exhibit extreme dispersion—from below 10% to above 95% on WN18RR, and from 0.8% to 84.3% across 237 FB15k-237 relations—and this heterogeneity persists across model architectures, consistent with the existence of “knowledge blind spots” that resist embedding-based prediction. Third, MRR and FDR produce conflicting model rankings: on WN18RR, ConvE trails TransE on MRR (0.469 vs. 0.560) yet achieves $2.8\times$ the BH rejection rate; the ranking divergence generalizes to FB15k-237 where the FDR ordering (RotatE > TransE > ConvE > ComplEx) disagrees with the MRR ordering (RotatE > ConvE > TransE > ComplEx).

These findings call into question the assumption that a single ranking metric adequately characterizes model quality. Our results suggest that the KG completion community would benefit from adopting FDR-based diagnostics alongside MRR and Hits@ K as a standard component of model evaluation. At minimum, we recommend reporting $\hat{\pi}_0$ and BH rejection counts at $\alpha = 0.05$, which together quantify the proportion of unpredictable test triples and the absolute

number of statistically reliable discoveries—information that ranking metrics alone cannot supply. The practical significance extends beyond benchmarking: for knowledge-based applications that consume KG completion output—drug repurposing, clinical decision support, financial risk monitoring—knowing which predictions are statistically reliable directly informs resource allocation.

The framework is compatible with major embedding paradigms, operates as a post-hoc protocol requiring no model modification, and completes in minutes on consumer hardware. We have released the complete source code, trained model checkpoints, and per-relation FDR statistics to facilitate community adoption.

Several limitations warrant further investigation. The effect of dependence among p -values on FDR control guarantees, while partially addressed through the Benjamini-Yekutieli procedure (see Sensitivity Analysis, §4.8), merits formal theoretical analysis for general KG embedding architectures. The current results are reported from single training runs; multi-seed evaluation would quantify the variance of FDR metrics and strengthen the reliability of comparative conclusions. Extending the analysis to larger benchmarks (YAGO3-10, Wikidata5M), additional model families (GNN-based, hyperbolic, LLM-enhanced), and real-world deployment case studies would further establish the generality and practical impact of FDR-based KG evaluation.

Acknowledgments

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Data Availability Statement

The WN18RR and FB15k-237 datasets used in this study are publicly available benchmark datasets. The complete source code, analysis scripts, trained

model checkpoints, and per-relation FDR statistics are openly available at <https://github.com/Jackxiaozihi>. No new data were generated for this study. All experiments were conducted with random seed 42; re-running with the provided code and seed will reproduce the exact results reported in this paper.

References

- [1] T. Shen, F. Zhang, J. Cheng, A comprehensive overview of knowledge graph completion, *Knowledge-Based Systems* 255 (2022) 109597. doi:10.1016/j.knosys.2022.109597.
- [2] R. Biswas, L.-A. Kaffee, M. Cochez, S. Dumbrava, T. E. Jendal, M. Lissandrini, V. Lopez, E. Loza Mencía, H. Paulheim, H. Sack, E. Kalemí Vakaj, G. de Melo, Knowledge graph embeddings: Open challenges and opportunities, *Transactions on Graph Data and Knowledge (TGDK)* 1 (1) (2023) 4:1–4:37. doi:10.4230/TGDK.1.1.4.
- [3] A. Bordes, N. Usunier, A. Garcia-Duran, J. Weston, O. Yakhnenko, Translating embeddings for modeling multi-relational data, in: *Advances in Neural Information Processing Systems (NeurIPS)*, 2013, pp. 2787–2795. doi:10.5555/2999792.2999923.
- [4] Z. Sun, Z.-H. Deng, J.-Y. Nie, J. Tang, RotatE: Knowledge graph embedding by relational rotation in complex space, in: *International Conference on Learning Representations (ICLR)*, 2019, pp. 1–20. doi:10.48550/arXiv.1902.10197.
- [5] T. Trouillon, J. Welbl, S. Riedel, É. Gaussier, G. Bouchard, Complex embeddings for simple link prediction, in: *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, 2016, pp. 2071–2080. doi:10.5555/3045390.3045609.
- [6] T. Dettmers, P. Minervini, P. Stenetorp, S. Riedel, Convolutional 2D knowledge graph embeddings, in: *Proceedings of the AAAI Conference*

- on Artificial Intelligence, 2018, pp. 1811–1818. doi:10.1609/aaai.v32i1.11573.
- [7] I. Balazević, C. Allen, T. M. Hospedales, TuckER: Tensor factorization for knowledge graph completion, in: Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP), 2019, pp. 5185–5194. doi:10.18653/v1/D19-1522.
 - [8] D. Q. Nguyen, T. D. Nguyen, D. Q. Nguyen, D. Phung, A novel embedding model for knowledge base completion based on convolutional neural network, in: Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT), 2018, pp. 327–333. doi:10.18653/v1/N18-2053.
 - [9] Z. Sun, J. Huang, Z.-H. Deng, Knowledge graph embedding with hierarchical structure via hyperbolic space, Knowledge-Based Systems 252 (2022) 108909. doi:10.1016/j.knosys.2022.108909.
 - [10] T.-B. Le, T. Tran, T. Tran, Knowledge graph embedding with the special orthogonal group in quaternion space for link prediction, Knowledge-Based Systems 264 (2023) 110400. doi:10.1016/j.knosys.2023.110400.
 - [11] Q. Wang, Z. Mao, B. Wang, L. Guo, Knowledge graph embedding: A survey of approaches and applications, IEEE Transactions on Knowledge and Data Engineering 29 (12) (2017) 2724–2743. doi:10.1109/TKDE.2017.2754499.
 - [12] A. Rossi, D. Barbosa, D. Firmani, A. Matinata, P. Merialdo, Knowledge graph embedding for link prediction: A comparative analysis, ACM Transactions on Knowledge Discovery from Data 15 (2) (2021) 1–49. doi:10.1145/3424672.
 - [13] X. Li, X. Fu, W. Hall, T. Tsiropas, Knowledge graph completion: A survey of datasets, methods, and evaluation metrics, ACM Computing Surveys 56 (3) (2023) 1–29. doi:10.1145/3626521.

- [14] X. Liu, F. Li, P. Shi, S. Singh, Understanding the limitations of knowledge graph embedding evaluation, *Transactions of the Association for Computational Linguistics* 10 (2022) 1303–1318. doi:10.1162/tac1_a_00524.
- [15] L. Jarnac, P. Monnin, M. Couceiro, A. Napoli, Y. Toussaint, Uncertainty management in the construction of knowledge graphs: A survey, *Transactions on Graph Data and Knowledge (TGDK)* 3 (1) (2025) 1–47. doi:10.4230/TGDK.3.1.1.
- [16] Y. Benjamini, Y. Hochberg, Controlling the false discovery rate: A practical and powerful approach to multiple testing, *Journal of the Royal Statistical Society: Series B (Methodological)* 57 (1) (1995) 289–300. doi:10.1111/j.2517-6161.1995.tb02031.x.
- [17] J. D. Storey, A direct approach to false discovery rates, *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 64 (3) (2002) 479–498. doi:10.1111/1467-9868.00346.
- [18] B. Efron, Large-scale simultaneous hypothesis testing: The choice of a null hypothesis, *Journal of the American Statistical Association* 99 (465) (2004) 96–104. doi:10.1198/016214504000000089.
- [19] W. Zhang, O. Şerban, J. Sun, Y. Guo, Conflict-aware multilingual knowledge graph completion, *Knowledge-Based Systems* 281 (2023) 111070. doi:10.1016/j.knosys.2023.111070.
- [20] B. Shang, Y. Zhao, J. Liu, Learnable convolutional attention network for knowledge graph completion, *Knowledge-Based Systems* 285 (2024) 111360. doi:10.1016/j.knosys.2023.111360.
- [21] M. Lu, Y. Li, J. Zhang, H. Ren, X. Zhang, Deep hyperbolic convolutional model for knowledge graph embedding, *Knowledge-Based Systems* 300 (2024) 112183. doi:10.1016/j.knosys.2024.112183.
- [22] D. Purohit, Y. Chudasama, A. Rivas, M.-E. Vidal, VANILLA: Validated knowledge graph completion—a normalization-based framework for in-

- tegrity, link prediction, and logical accuracy, *Knowledge-Based Systems* 318 (2025) 113939. doi:10.1016/j.knosys.2025.113939.
- [23] Y. Fang, X. Liu, W. Lu, W. Pedrycz, Q. Lang, J. Yang, Knowledge graph completion with low-dimensional gated hierarchical hyperbolic embedding, *Knowledge-Based Systems* 309 (2025) 112804. doi:10.1016/j.knosys.2024.112804.
- [24] M. Ali, M. Berrendorf, C. T. Hoyt, L. Vermue, M. Galkin, S. Sharifzadeh, A. Fischer, V. Tresp, J. Lehmann, PyKEEN 1.0: A Python library for training and evaluating knowledge graph embeddings, *Journal of Machine Learning Research* 22 (82) (2021) 1–6. doi:10.48550/arXiv.2007.14153.
- [25] S. Vashishth, S. Sanyal, V. Nitin, P. Talukdar, Composition-based multi-relational graph convolutional networks, in: *International Conference on Learning Representations (ICLR)*, 2020, pp. 1–19. doi:10.48550/arXiv.1911.03082.
- [26] Z. Zhu, Z. Zhang, L.-P. Xhonneux, J. Tang, Neural Bellman-Ford networks: A general graph neural network framework for link prediction, in: *Advances in Neural Information Processing Systems (NeurIPS)*, 2021, pp. 29476–29490. doi:10.5555/3540261.3542517.
- [27] Z. Gao, H. Wang, Z. Li, A. Sun, E. Chen, Unifying knowledge graph embedding and retrieval with large language models, in: *Proceedings of the ACM Web Conference (WWW)*, 2025, pp. 2155–2166. doi:10.1145/3696410.3714587.
- [28] N. Reimers, I. Gurevych, Reporting score distributions makes a difference: Performance study of LSTM-networks for sequence tagging, in: *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2017, pp. 338–348. doi:10.18653/v1/D17-1035.
- [29] T. Safavi, D. Koutra, CoDEx: A comprehensive knowledge graph completion benchmark, in: *Proceedings of the Conference on Empirical Meth-*

- ods in Natural Language Processing (EMNLP), 2020, pp. 8328–8350. doi:10.18653/v1/2020.emnlp-main.669.
- [30] X. Wang, T. Chen, Y. Du, Z. Li, H. Ji, Revisiting knowledge graph completion evaluation, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2023, pp. 10159–10167. doi:10.1609/aaai.v37i8.26208.
 - [31] H. Shomer, W. Jin, W. Wang, J. Tang, Toward degree bias in embedding-based knowledge graph completion, in: Proceedings of the ACM Web Conference (WWW), 2023, pp. 1022–1033. doi:10.1145/3543507.3583544.
 - [32] O. Lloyd, T. R. Gaunt, Y. Liu, Assessing the effects of hyperparameters on knowledge graph embedding quality, *Journal of Big Data* 10 (2023) 72. doi:10.1186/s40537-023-00732-5.
 - [33] P. Tabacof, L. Costabello, Probability calibration for knowledge graph embedding models, in: Proceedings of the 3rd Alberto Mendelzon International Workshop on Foundations of Data Management (AMW), 2020, pp. 1–12, available at: <https://ceur-ws.org/Vol-2578/>.
 - [34] S. He, K. Liu, G. Ji, J. Zhao, Learning to represent knowledge graphs with Gaussian embedding, in: Proceedings of the 24th ACM International Conference on Information and Knowledge Management (CIKM), 2015, pp. 623–632. doi:10.1145/2806416.2806502.
 - [35] T. Wu, S. Zhu, J. Wang, N. Xu, G. Qi, H. Wang, Uncertain knowledge graph completion via semi-supervised confidence distribution learning, in: Advances in Neural Information Processing Systems (NeurIPS), 2025, pp. 1–18. doi:10.48550/arXiv.2510.16601.
 - [36] A. Marandon, Conformal link prediction for false discovery rate control, *TEST* 33 (4) (2024) 1062–1083. doi:10.1007/s11749-024-00934-w.
 - [37] Y. Zhu, D. Hernandez, Y. He, Z. Ding, B. Xiong, E. Kharlamov, S. Staab, Predicate-conditional conformalized answer sets for knowledge graph em-

- beddings, in: Findings of the Association for Computational Linguistics (ACL), 2025, pp. 4145–4167. doi:10.18653/v1/2025.findings-acl.215.
- [38] Y. Zhu, J. Wu, Y. Wang, H. Zhou, J. Chen, E. Kharlamov, S. Staab, Certainty in uncertainty: Reasoning over uncertain knowledge graphs with statistical guarantees, in: Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025, pp. 8730–8752. doi:10.18653/v1/2025.emnlp-main.441.
- [39] B. Ni, Y. Wang, L. Cheng, E. Blasch, T. Derr, Towards trustworthy knowledge graph reasoning: An uncertainty aware perspective, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2025, pp. 12417–12425. doi:10.1609/aaai.v39i12.33353.
- [40] Y. Zhu, N. Potyka, M. Nayyeri, B. Xiong, Y. He, E. Kharlamov, S. Staab, Predictive multiplicity of knowledge graph embeddings in link prediction, in: Findings of the Association for Computational Linguistics: EMNLP 2024, 2024, pp. 334–354. doi:10.18653/v1/2024.findings-emnlp.19.
- [41] Z. Sun, S. Vashishth, S. Sanyal, P. Talukdar, Y. Yang, A re-evaluation of knowledge graph completion methods, in: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL), 2020, pp. 5516–5522. doi:10.18653/v1/2020.acl-main.489.
- [42] L. Schwertmann, M. P. Kannan Ravi, G. de Melo, Model-agnostic bias measurement in link prediction, in: Findings of the Association for Computational Linguistics: EACL 2023, 2023, pp. 1626–1634. doi:10.18653/v1/2023.findings-eacl.121.
- [43] J. D. Storey, R. Tibshirani, Statistical significance for genomewide studies, Proceedings of the National Academy of Sciences 100 (16) (2003) 9440–9445. doi:10.1073/pnas.1530509100.
- [44] B. Efron, Size, power and false discovery rates, The Annals of Statistics 35 (4) (2007) 1351–1377. doi:10.1214/009053606000001460.

- [45] J. Demšar, Statistical comparisons of classifiers over multiple data sets, *Journal of Machine Learning Research* 7 (2006) 1–30. doi:10.5555/1248547.1248548.
- [46] J. Taylor, R. J. Tibshirani, Statistical learning and selective inference, *Proceedings of the National Academy of Sciences* 112 (25) (2015) 7629–7634. doi:10.1073/pnas.1507583112.
- [47] J. D. Lee, D. L. Sun, Y. Sun, J. E. Taylor, Exact post-selection inference, with application to the lasso, *The Annals of Statistics* 44 (3) (2016) 907–927. doi:10.1214/15-AOS1371.
- [48] R. F. Barber, E. J. Candès, Controlling the false discovery rate via knock-offs, *The Annals of Statistics* 43 (5) (2015) 2055–2085. doi:10.1214/15-AOS1337.
- [49] E. Candès, Y. Fan, L. Janson, J. Lv, Panning for gold: ‘Model-X’ knockoffs for high dimensional controlled variable selection, *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 80 (3) (2018) 551–577. doi:10.1111/rssb.12265.
- [50] A. N. Angelopoulos, S. Bates, Conformal prediction: A gentle introduction, *Foundations and Trends in Machine Learning* 16 (4) (2023) 494–591. doi:10.1561/22000000101.
- [51] A. N. Angelopoulos, R. F. Barber, S. Bates, Theoretical foundations of conformal prediction, arXiv preprint arXiv:2411.11824Forthcoming book, Cambridge University Press (2024). doi:10.48550/arXiv.2411.11824.
- [52] S. Bates, T. Hastie, R. Tibshirani, Cross-validation: What does it estimate and how well does it do it?, *Journal of the American Statistical Association* 119 (546) (2024) 1434–1445. doi:10.1080/01621459.2023.2197686.
- [53] G. Durand, R. Perier, A. Marandon, FDR control and FDP bounds for conformal link prediction, arXiv preprint arXiv:2404.02542 (2024). doi:10.48550/arXiv.2404.02542.

- [54] Y. Benjamini, D. Yekutieli, The control of the false discovery rate in multiple testing under dependency, *The Annals of Statistics* 29 (4) (2001) 1165–1188. doi:10.1214/aos/1013699998.
- [55] A. C. Davison, D. V. Hinkley, *Bootstrap Methods and Their Application*, Cambridge University Press, 1997. doi:10.1017/CB09780511802843.
- [56] K. Toutanova, D. Chen, P. Pantel, H. Poon, P. Choudhury, M. Gamon, Representing text for joint embedding of text and knowledge bases, in: *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2015, pp. 1499–1509. doi:10.18653/v1/D15-1174.
- [57] D. Krompaß, M. Nickel, V. Tresp, Type-constrained representation learning in knowledge graphs, in: *Proceedings of the 14th International Semantic Web Conference (ISWC)*, 2015, pp. 253–269. doi:10.1007/978-3-319-25007-6_15.
- [58] Y. Benjamini, A. M. Krieger, D. Yekutieli, Adaptive linear step-up procedures that control the false discovery rate, *Biometrika* 93 (3) (2006) 491–507. doi:10.1093/biomet/93.3.491.

Algorithm 1 FDR-Based KG Completion Evaluation Protocol

Require: Trained KG embedding model with scoring function $\hat{g}$; Test set

$\mathcal{T}_{\text{test}} = \{(h_i, r_i, t_i^*)\}_{i=1}^m$; Entity set $\mathcal{E}$; number of negative samples K ; FDR level α

Ensure: p -values $\{\hat{p}_i\}_{i=1}^m$; BH rejections $R(\alpha)$; Storey $\hat{\pi}_0$, q -values; local FDR

```
1: // Phase 1:  $p$ -value construction
2: for  $i = 1$  to  $m$  do
3:   Sample  $K$  negative tails  $\{t_k^{(i)}\}_{k=1}^K \subset \mathcal{E}$  (filtered)
4:   Compute  $s_i^* = \hat{g}(h_i, r_i, t_i^*)$  ▷ True triple score
5:   for  $k = 1$  to  $K$  do
6:     Compute  $s_{i,k} = \hat{g}(h_i, r_i, t_k^{(i)})$  ▷ Negative triple score
7:   end for
8:    $\hat{p}_i \leftarrow \frac{|\{k: s_{i,k} \geq s_i^*\}| + 1}{K + 1}$  ▷ Eq. 2
9: end for
10: // Phase 2: Benjamini-Hochberg FDR control
11: Sort  $\{\hat{p}_i\}_{i=1}^m$  ascending:  $\hat{p}_{(1)} \leq \dots \leq \hat{p}_{(m)}$ 
12:  $k^* \leftarrow \max\{k : \hat{p}_{(k)} \leq \frac{k}{m} \cdot \alpha\}$ 
13:  $R(\alpha) \leftarrow k^*$  ▷ Number of discoveries at FDR level  $\alpha$ 
14: // Phase 3: Storey's  $q$ -value and  $\hat{\pi}_0$  estimation
15: for  $\lambda \in \Lambda$  do ▷ Grid of tuning parameters, e.g.,  $\Lambda = \{0.1, \dots, 0.9\}$ 
16:    $\hat{\pi}_0(\lambda) \leftarrow \frac{|\{i: \hat{p}_i > \lambda\}|}{m \cdot (1 - \lambda)}$  ▷ Eq. 4
17: end for
18:  $\hat{\pi}_0 \leftarrow \text{smooth}\{\hat{\pi}_0(\lambda)\}_{\lambda \rightarrow 1}$  ▷ Cubic spline extrapolation
19: Compute  $q$ -values from  $\{\hat{p}_i\}$  and  $\hat{\pi}_0$ 
20: // Phase 4: Efron's local FDR
21:  $z_i \leftarrow \Phi^{-1}(1 - \hat{p}_i)$  for all  $i$  ▷ Eq. 6
22: Estimate mixture density  $f(z)$  via kernel density estimation
23:  $\text{fdr}(z_i) \leftarrow \frac{\hat{\pi}_0 \cdot \phi(z_i)}{f(z_i)}$  for all  $i$  ▷ Eq. 8
24: return  $\{\hat{p}_i\}, R(\alpha), \hat{\pi}_0, \{q_i\}, \{\text{fdr}(z_i)\}$ 
```
